Uchaaran

A playful path to confident Gurbani pronunciation

Team 12 | Partner: Khalis Foundation
Avi Oberoi, Gurjaap Kalra, Mohkm Singh, Gurasees Bajaj

Selected problem: Help children build a lasting Gurbani practice through pronunciation support and meaningful progress.

The problem, in our words

Children learning Gurbani can struggle to read Gurmukhi and know whether they are pronouncing a line correctly. Parents want to help, but may not have time or access to an ustaad for everyday practice.

What we built

A working browser prototype inspired by Sundar Gutka: 38 Japji Sahib pauris across six map regions, guided practice, challenge results and character rewards. Audio and saved progress work; pronunciation scores and coaching are simulated.

Who uses it and how it helps

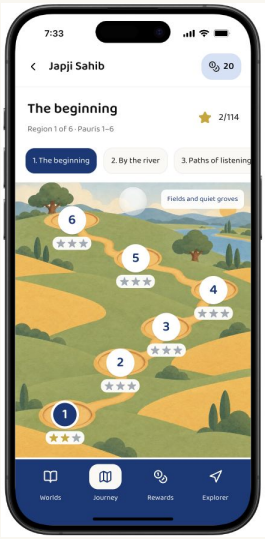
For children in grades 6 to 8, with parent support.

Choose a path. Pick Prem Singh or Sehaj Kaur. Open a pauri on the map and preview its connected visual story.

Practice with support. Read Gurmukhi, Roman-letter Punjabi and English meaning. Listen to reference audio; revisit difficult lines.

See progress. Review word-level feedback. Earn stars and Nanak coins for clothing, belongings and character actions.

For Khalis Foundation, this tests a learning journey within Sundar Gutka. Success means children return and improve against an ustaad-reviewed standard.



Tech tools we used

JavaScript, React Native Web and Vite; BaniDB text and Khalis audio; Codex, Paper, ChatGPT Sites and AI-generated illustrations.

Links

Try the live prototype | Review the code on GitHub

Next steps and asks

Before the final submission, polish the full practice-to-rewards flow, test usability and advance the React Native voice-following integration. We welcome mentor help with speech alignment and access to test families.

Working with the nonprofit

Khalis Foundation gave the project a clear purpose: helping children connect with Gurbani. Its books, audio and Sundar Gutka app grounded our designs in familiar resources and kept learning at the center of rewards.

Hackathon feedback

Building around a real nonprofit need gave Build Day a clear focus. More time for user testing and targeted speech-technology mentorship would help turn the prototype into a dependable learning tool.